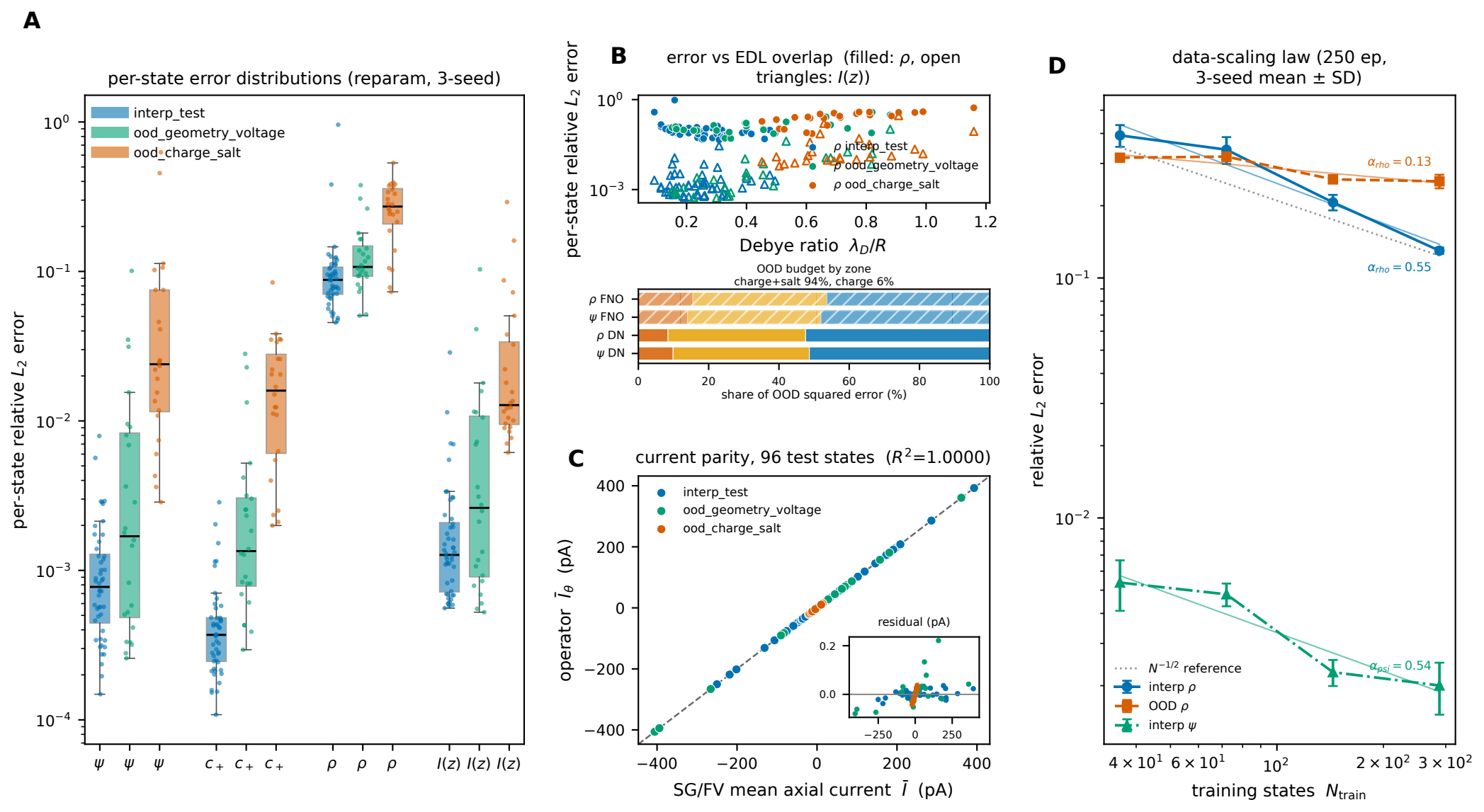

A
per-state error distributions (reparam, 3-seed)
interp_test
ood_geometry_voltage
ood_charge_salt
per-state relative L_2 error
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
ψ ψ ψ c_+ c_+ c_+ ρ ρ ρ $I(z)$ $I(z)$ $I(z)$
B
error vs EDL overlap (filled: ρ, open triangles: $I(z)$)
per-state relative L_2 error
ρ interp_test
ρ ood_geometry_voltage
ρ ood_charge_salt
Debye ratio λ_D/R
OOD budget by zone
charge+salt 94%, charge 6%
ρ FNO
ψ FNO
ρ DN
ψ DN
share of OOD squared error (%)
C
current parity, 96 test states (R^2=1.0000)
interp_test
ood_geometry_voltage
ood_charge_salt
operator $\bar{I}_\theta$ (pA)
residual (pA)
SG/FV mean axial current $\bar{I}$ (pA)
D
data-scaling law (250 ep, 3-seed mean ± SD)
relative L_2 error
$\alpha_{rho} = 0.13$
$\alpha_{rho} = 0.55$
$\alpha_{psi} = 0.54$
$N^{-1/2}$ reference
interp ρ
OOD ρ
interp ψ
training states N_{train}